

A MULTI-LAYER, SELF-ALIGNING HYDROGEL MICROMOLDING PROCESS OFFERING A FABRICATION ROUTE TO PERFUSABLE 3D IN-VITRO MICROVASCULATURE

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ABSTRACT

We introduce a versatile method to create multi-layered, cell-laden hydrogel microstructures with coaxial geometries and heterogeneous elastic moduli. The technique can be used to build *in-vitro* vascular constructs that are completely surrounded by material with physiologically realistic mechanical stiffnesses. The composition and mechanical properties of the agarose–collagen (AC) composite hydrogels have been tuned for optimal cell viability and biological performance.

KEYWORDS: Hydrogel, *in-vitro* vascular construct, elastic moduli, agarose-collagen composite

INTRODUCTION

Existing *in-vitro* models of vasculature expose endothelia to materials that are far stiffer than they encounter in the body: for example, microfluidic devices [1],[2], even if injected with extra-cellular matrix (ECM), frequently incorporate elastomeric surfaces that are 100–1000 times stiffer than typical ECM; meanwhile, polystyrene trans-well membranes [2],[3] are around 10^6 times higher in elastic modulus. Our technique eliminates those rigid polymeric surfaces from the vicinity of the cells and allows layers of multiple cell types to be defined with tailored ECM composition and stiffness, and in direct contact with each other.

EXPERIMENTAL

The process (Fig 1) uses a sequence of molded thermal gelation steps to build the microstructure. Each mold is produced by an Objet Connex-260 3D inkjet printer with a nominal resolution of 16 μm . The material to be molded is heated, dispensed, deformed by the mold, and then allowed to gel thermally as it cools (Fig 1a–b). The molded material can be, for example, an aqueous solution of agarose or collagen, or a mixture of materials. Cells can be dispersed throughout the layer, enabling 3D culture (Fig 2c), or seeded onto the hydrogel surface after the layer is complete (Fig 2d). The composition and cell content of each layer are independently controllable. In order to achieve multi-layer coaxial structures, one molding step is followed by another with the same layout but smaller-diameter protrusions, so that a half-cylindrical shell of hydrogel is cast into the intermediate space (Fig 1c–d). This sequence can be extended to multiple layers at different radial locations (Fig 1d). Layer-to-layer registration is enabled by the self-centering action of mold protrusions within previously molded hydrogel features (Fig 1d). The un-gelled solution lubricates the interface. Posts printed onto the molds further enhance registration, and enable the two halves of the hydrogel construct to interlock and form fully enclosed, coaxial structures (Fig 1e–f).

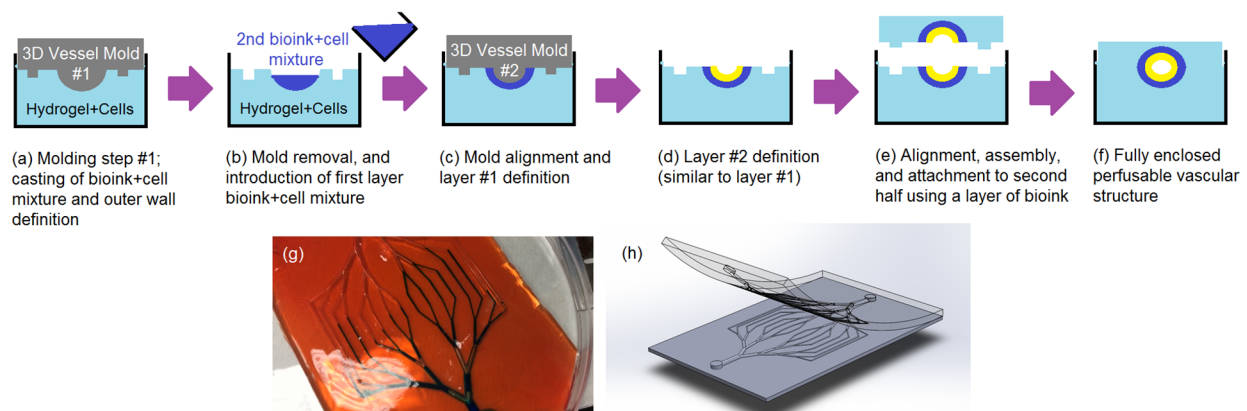


Figure 1: Fabrication method: (a–f) process flow; (g) flow in the assembled perfusable network; (h) illustration of 3d printed mold design to produce that network.

RESULTS AND DISCUSSION

We have demonstrated co-axial geometries with diameters ranging from 200–2000 μm and layer thicknesses as small as 50–200 μm in AC hydrogels (Fig 2a). Coaxial geometries with such fine feature sizes are beyond the capabilities of most bioprinting techniques [4]. The triple-layer construct of Fig 2b shows such patterning capability, having an inner shell and outer region populated with SY5Y glial cells, and an intermediate layer without cells. Agarose makes the hydrogel moldable with tunable elastic modulus in the range of 1–10 kPa (Fig 3), while the presence of collagen ensures acceptable viability (Fig 4). This prototype *in-vitro* model resembles vascular geometries and opens the way for intricate multi-luminal blood vessels to be fabricated.

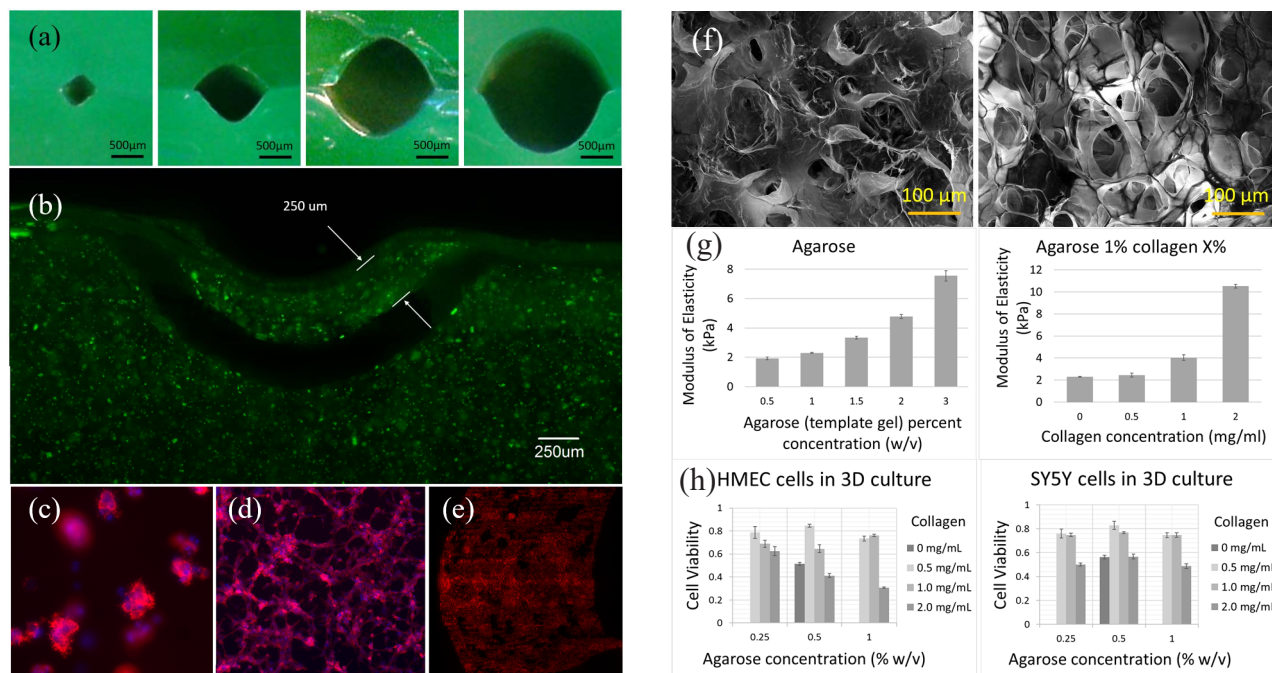


Figure 2: Fabricated hydrogel constructs; mechanical properties and biocompatibility: (a) cross-sections of vessels with diameters varying from 250–1500 μm (b) tri-layered construct with an inner diameter of 500 μm (c) expansion and colonization of SY5Y glial cells in 3D culture mode (d) spreading of HMEC cells on poly-d-lysine coated agarose-collagen surfaces in 2D culture mode (e) confocal image of 3D cellular lumen following the topography of the molded hydrogel layer (f) SEM images of gel structure indicating the added structural support of collagen fiber bundles as an ECM backbone; (g) stiffness values matching those of *in-vivo* cerebral and vascular tissue. (h) Viability analysis results for SY5Y and HMEC cells in 3D culture after seven days *in-vitro*. Low cell viability and loss of cell-adhesiveness at zero collagen and enhanced viability at specific collagen concentrations (Red stain: actin, blue/green stain: nuclei. Error bars show standard deviation among four trials)

CONCLUSION

We report the successful fabrication of a 3D multi-layered endothelial lumen with configurable layer thickness and vessel diameter. The hydrogel composition has been optimized for cell viability and structural performance.

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